Nonlinear Calvo Pricing, the Reset-Price Floor, and the Paradox of Flexibility at the Zero Lower Bound

Minseong Kim

November 4, 2026

Abstract

In the log-linearized New Keynesian model, more flexible prices deepen a slump at the zero lower bound (the paradox of flexibility). We solve the exact nonlinear Calvo model as prices become more flexible under strict price stability after the trap, whereas earlier nonlinear work varied shock persistence (Eggertsson and Singh 2019). The exact Calvo price index bounds the reset price from below, and this bound approaches the price level as prices become more flexible. It turns out that this bound makes the deflationary equilibrium cease to exist once the Calvo parameter falls below a positive threshold. With an economy-wide labor market, a two-year trap and a large negative natural rate, this threshold lies slightly above the conventional calibration. The Rotemberg model, with the same log-linear approximation, keeps its equilibrium at far more flexible prices in short traps, but its adjustment costs use up nearly all output in long traps.

JEL Classification: E31, E43, E52

Keywords: Zero Lower Bound, Paradox of Flexibility, Calvo Pricing, Rotemberg Pricing, Price Dispersion, Existence of Equilibrium, New Keynesian Macroeconomics

1 Introduction

Krugman (1998) revived the analysis of the liquidity trap in a modern intertemporal model and argued that monetary policy can escape the trap by credibly committing to future inflation. Much of the New Keynesian analysis of the zero lower bound follows Eggertsson and Woodford (2003). In their model a temporary shock pushes the natural rate of interest below zero, with the nominal rate unable to track the natural rate. The real rate therefore stays too high, and the economy goes through deflation and a large output contraction. They also showed that a credible commitment to keep rates low after the shock has passed can greatly reduce the damage.

Later papers derived several unusual predictions from the same model. Eggertsson (2011) found large government-spending multipliers at the zero bound. Eggertsson and Krugman (2012) described a paradox of flexibility, which is the focus of this paper: more flexible prices deepen the slump because faster deflation raises the real rate further.

Werning (2012) derives the paradox of flexibility analytically. He uses a deterministic continuous-time version of the log-linearized model in which the natural rate is negative for a known period T and the central bank keeps inflation and the output gap at zero from T on. He then solves the model backward from T and shows that the slump grows without bound as prices become fully flexible. He also notes that this flexible-price limit is not the same as the flexible-price equilibrium.

Later work has questioned these predictions on two grounds: equilibrium selection and the log-linear approximation. What happens during the trap depends on what households and firms expect after T, and this expectation is fixed by the terminal condition. Cochrane (2017) shows that other terminal conditions are equally consistent with the model and argues that the paradoxes (including the paradox of flexibility) are artifacts of the standard equilibrium selection. Also note that under an active Taylor rule the zero bound creates a second deflationary steady state (Benhabib, Schmitt-Grohé and Uribe, 2001), so selection issues are not limited to the terminal condition.

Most of the papers discussed so far also use the log-linearized model, which approximates each equilibrium condition around the zero-inflation steady state. This approximation is least reliable for very large deflations, which arise in the paradox of flexibility. For example, Boneva, Braun and Waki (2016) solve a nonlinear model with Rotemberg pricing and find that log-

linearized solutions at the zero bound can be misleading. Fernández-Villaverde et al. (2015) and Gavin et al. (2015) go further and solve stochastic New Keynesian models with an occasionally binding zero bound by global methods, which keeps the nonlinearity that the bound creates.

The way prices are set raises a further nonlinear issue. Under Calvo pricing a random fraction $1-\theta$ of firms resets prices each period, so the nonlinear model contains price dispersion, which is a source of nonlinear effects. Identical firms charge different prices because they reset at different times. Dispersion misallocates production and raises the labor needed per unit of output. It is second-order in inflation and so vanishes from the log-linearized model.

In contrast to the Calvo model, dispersion is absent from the Rotemberg model, in which every firm adjusts its price each period but pays a cost to do so. Around the zero-inflation steady state, this adjustment cost is second-order in inflation just like Calvo price dispersion, and log-linearization therefore discards it. The Rotemberg cost parameter can then be chosen so that the log-linearized Rotemberg model takes the same form as the log-linearized Calvo model. The equivalence is specific to zero trend inflation: around a steady state with positive trend inflation, the two models differ even to first order (Ascari and Rossi, 2012). Even with zero trend inflation, Miao and Ngo (2021) find that the nonlinear versions of the two models behave quite differently at the zero bound, with more deflation and deeper recessions under Calvo pricing. They attribute the difference mainly to the Rotemberg adjustment costs, which enter the resource constraint, rather than to price dispersion.

Eggertsson and Singh (2019) look at dispersion more directly. They solve the nonlinear Calvo model under a two-state Markov shock without approximation and show that the role of dispersion depends on how the labor market is modeled. When each firm hires from a separate labor market, dispersion does not affect aggregate output and inflation, and the log-linearized solution is close to the exact one.

None of these papers really solves the exact nonlinear model as prices become more flexible under the standard terminal condition, and this paper intends to fill this gap. We do this in a discrete-time version of the deterministic setup of Werning (2012) and ask whether the paradox arises only from the log-linear approximation or also from the full nonlinear model. We show that the paradox is also a result of the full nonlinear model, although there it appears mainly as a problem of equilibrium existence (or of equilibrium selection) as prices become flexible. The problem is more severe in the nonlinear Calvo model than in its log-linear approximation.

Our argument proceeds in three steps. First, we identify a property of the Calvo price index

that log-linearization discards. The price level gravitates toward the lowest prices because consumers substitute toward cheaper goods, and during a deflation the lowest prices are the reset prices of firms not bound by price rigidity in that period. The reset price therefore cannot fall far below the price level: $P_t^*/P_t > (1-\theta)^{1/(\epsilon-1)}$, where ϵ is the elasticity of substitution (Lemma 2). We call this lower bound the reset-price floor. The reset-price floor approaches one as $\theta \to 0$, and the log-linearized model has no reset-price floor.

Second, we show that the reset-price floor rules out the standard terminal condition once prices are flexible enough. In the last period of the trap, resetting firms have to cut prices by an amount that depends on the shock but not on θ. The reset-price floor forbids this cut once θ falls below a critical value θ^*, and no deflationary equilibrium then exists. We call θ^* the existence threshold and use this term interchangeably with existence boundary. For the last trap period we prove this with firm-specific labor markets (Proposition 1). With an economy-wide labor market we demonstrate, via simulating over intervals of θ, that no such equilibrium exists below the existence threshold. Proposition 2 proves non-existence only for sufficiently flexible prices and locates the threshold only to first order. We solve the exact model numerically to show that both results extend to longer traps. With firm-specific labor markets and under Rotemberg pricing we also prove that a longer trap never lowers the existence threshold (Proposition 5). In the numerical solutions the thresholds rise quickly with the length of the trap. For a two-year trap with a large negative natural rate and an economy-wide labor market, θ^* lies slightly above the conventional calibration (Table 1). With firm-specific labor markets the thresholds are lower.

Deflation grows without bound as θ falls toward θ^*. By contrast, in the deterministic log-linearized model this happens only in the limit of fully flexible prices. Away from the existence threshold the exact and log-linearized paths are nearly identical, as Eggertsson and Singh (2019) found.

Price dispersion surprisingly matters little for the existence boundary, although it is the most obvious nonlinear effect in the Calvo model. Strict price stability after T remains feasible when dispersion is inherited from the trap. It only requires a small second-order output gap while dispersion decays (Lemma 1). Deflation also generates little dispersion near the flexible-price limit because firms stuck at high prices sell almost nothing (Lemma 3).

Third, we compare the Calvo model with the Rotemberg model, which has the same log-linear approximation but no reset-price floor. The Rotemberg equilibrium is limited instead by

the resources used up in changing prices. Miao and Ngo (2021) identify these resource costs as the main source of the difference between the two models, and Eggertsson and Singh (2019) also point to the large resource costs of price adjustment in the Rotemberg model. In a one-period trap the Rotemberg equilibrium exists at far more flexible prices than the Calvo equilibrium (Proposition 4 and Table 1). In longer traps the equilibrium exists only with adjustment costs that absorb nearly all output. In a two-year trap these costs exceed half of output at some date already at degrees of price stickiness close to the economy-wide Calvo existence threshold (Table 1).

Our existence results are consistent with the equilibrium-selection argument of Cochrane (2017). The non-existence we find concerns the standard terminal condition. Other terminal conditions (for example one in which the price level jumps up when the shock ends) still yield equilibria. In the exact Calvo model the standard selection fails before prices become fully flexible, and deflation grows without bound as that point is approached.

1.1 Comparison with Eggertsson and Singh (2019)

Eggertsson and Singh (2019) also find that the exact Calvo model may not have an equilibrium when the zero lower bound is binding. However, their results are based on stochastic shocks, and although they state, without further analysis, that the nonlinear solution also converges to a finite limit as prices become more flexible, they trace the loss of an equilibrium only along the shock-persistence probability μ, which is necessary for their results. Our model is deterministic, and its loss of equilibrium depends on the Calvo parameter θ (or its Rotemberg equivalent). The analysis in this paper therefore further establishes that the paradox of flexibility has to be re-examined: loss of an equilibrium comes before the paradox of flexibility hits.

In their model the trap ends in each period with probability $1-\mu$, after which the shock stays at its steady-state value. With firm-specific labor markets, and with output and inflation constant while the trap lasts, the model reduces to an aggregate demand curve and an aggregate supply curve in these two variables. In the log-linearized model both curves are straight lines, and they become parallel as μ rises. At a critical persistence $\bar{\mu}$ output and inflation diverge, and for $\mu > \bar{\mu}$ the log-linearized model is indeterminate. The critical persistence is where condition C2 of Eggertsson (2011, Proposition 3) holds with equality. In our notation the condition reads $(1-\mu)(1-\beta\mu) - \kappa\mu/\sigma > 0$, where κ is the slope of the log-linear Phillips curve, and it requires the aggregate demand curve to be steeper than the aggregate supply curve (Eggertsson and

Singh, 2019, eq. 10). In the nonlinear model the aggregate supply curve is convex and becomes tangent to the aggregate demand curve as μ rises, at finite output and inflation. For higher persistence the two curves no longer intersect, and the nonlinear model has no equilibrium of this form. Eggertsson and Singh (2019) call the critical persistence the bifurcation point in both models. They establish the non-existence numerically at their calibration and then study equilibria with persistence below the critical value, where they also show that greater price flexibility deepens the slump in the exact model.

Our trap instead has a known length T, and we evaluate existence across values of θ with T fixed, which Eggertsson and Singh (2019) do not do. In this deterministic setting the log-linearized model has a solution for every $\theta \in (0, 1)$, and its slump becomes unbounded only as $\theta \to 0$. In their stochastic log-linearized model, greater price flexibility steepens the Phillips curve and lowers the critical persistence $\bar{\mu}$, because the slope of the Phillips curve enters condition C2. Persistence and flexibility then jointly determine existence, and a known exit date removes this interaction. In our exact deterministic model the existence boundary in θ comes from the reset-price floor rather than from a tangency of two curves, and deflation grows without bound as the existence boundary is approached. For the last trap period we prove that no deflationary equilibrium exists once prices are flexible enough, under both labor-market structures. We also prove that a longer trap never lowers the existence threshold with firm-specific labor markets and under Rotemberg pricing, and we solve the exact model numerically for longer traps and for an economy-wide labor market. With an economy-wide labor market price dispersion is a state variable, so the reduction of the model to two curves is not available.

1.2 Scope and organization

The analysis is deterministic, with a known exit date and a zero-rate peg during the trap. We assume zero trend inflation, no indexation and a linear technology, and the numerical results use one baseline calibration ($\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$ per quarter) with one-parameter-at-a-time variations. A stochastic exit date with an occasionally binding zero lower bound would require global solution methods, and we leave it for future work.

Section 2 presents the model. Section 3 establishes the properties of Calvo pricing used throughout. Section 4 derives the analytical results for the last trap period, and Section 5 studies longer traps, with a monotonicity result in the trap length and numerical solutions of the exact model. Section 6 discusses equilibrium selection, and Section 7 concludes. Derivations,

proofs and numerical methods are in the appendix.

2 Model

2.1 Households

A representative household maximizes

$$\sum_{t=0}^{\infty}\left(\prod_{j=0}^{t-1}\beta_j\right)\left(\frac{C_t^{1-\sigma}}{1-\sigma}-\chi\frac{N_t^{1+\varphi}}{1+\varphi}\right), \tag{1}$$

where β_t is the discount factor between t and $t+1$. Consumption C_t is a CES aggregate of varieties with elasticity $\epsilon > 1$, and P_t is the associated price index. With one-period nominal bonds paying i_t, the Euler equation is

$$C_t^{-\sigma} = \beta_t\, C_{t+1}^{-\sigma}\,\frac{1+i_t}{\Pi_{t+1}}, \qquad \Pi_{t+1} \equiv P_{t+1}/P_t. \tag{2}$$

Holding output constant, the natural real rate is $r_t^n = 1/\beta_t - 1$. We normalize technology to one and set χ so that steady-state output satisfies $Y^{\sigma+\varphi} = (\epsilon-1)/\epsilon$, i.e. steady-state real marginal cost equals $MC \equiv (\epsilon-1)/\epsilon$. Let $\zeta \equiv \sigma + \varphi$.

2.2 Firms and labor markets

Firm i produces $Y_{i,t} = N_{i,t}$ and faces demand $Y_{i,t} = (P_{i,t}/P_t)^{-\epsilon}Y_t$. Goods market clearing is $C_t = Y_t$ under Calvo pricing. We consider two labor-market structures.

Economy-wide labor market. All firms hire from a single market at real wage $W_t/P_t = \chi N_t^{\varphi} C_t^{\sigma}$. Aggregating labor demand gives $N_t = S_t Y_t$, where

$$S_t \equiv \int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} di \tag{3}$$

is price dispersion (Yun, 1996, p. 355). Real marginal cost, common to all firms, is

$$MC_t = \chi Y_t^{\zeta} S_t^{\varphi}, \qquad \frac{MC_t}{MC} = e^{\zeta x_t} S_t^{\varphi}, \tag{4}$$

where $x_t \equiv \log(Y_t/Y)$ is the log deviation of output from its steady-state level Y. There are no technology shocks, so output under flexible prices always equals Y, and x_t is also the output

gap.

Firm-specific (industry-specific) labor markets. Each firm hires from a separate labor market, taking the wage as given, with $W_{i,t}/P_t = \chi N_{i,t}^{\varphi} C_t^{\sigma}$ (Woodford, 2003, ch. 3, pp. 147–148; Eggertsson and Singh, 2019). Firm i's real marginal cost is then $MC_t^s \left(P_{i,t}/P_t\right)^{-\epsilon\varphi}$, where $MC_t^s \equiv \chi Y_t^{\zeta}$ is the marginal cost of a firm charging the average price. Price dispersion does not enter MC_t^s.

2.3 Calvo pricing

In each period, a randomly drawn fraction $1-\theta$ of firms resets its price to P_t^*, while the remaining firms keep their previous prices. The price index and dispersion then evolve as follows (Appendix A.3):

$$1 = (1-\theta) Z_t^{1-\epsilon} + \theta \Pi_t^{\epsilon-1}, \qquad Z_t \equiv P_t^*/P_t, \tag{5}$$

$$S_t = (1-\theta) Z_t^{-\epsilon} + \theta \Pi_l^{\epsilon} S_{t-1}. \tag{6}$$

With an economy-wide labor market, the optimal reset price satisfies

$$Z_t = \frac{\epsilon}{\epsilon - 1} \frac{V_{1,t}}{V_{2,t}}, \quad V_{1,t} = Y_t^{1-\sigma} MC_t + \beta_t \theta \, \Pi_{t+1}^{\epsilon} V_{1,t+1}, \quad V_{2,t} = Y_t^{1-\sigma} + \beta_t \theta \, \Pi_{t+1}^{\epsilon-1} V_{2,t+1}. \tag{7}$$

With firm-specific labor markets, a resetting firm takes into account that a higher relative price lowers its output and therefore its marginal cost. Its optimal reset price then satisfies (Appendix A.5)

$$Z_t^{1+\epsilon\varphi} = \frac{\epsilon}{\epsilon - 1} \frac{V_{1,t}^I}{V_{2,t}}, \qquad V_{1,t}^I = Y_t^{1-\sigma} MC_t^s + \beta_t \theta \, \Pi_{t+1}^{\epsilon(1+\varphi)} V_{1,t+1}^I, \tag{8}$$

with $V_{2,t}$ as in Equation (7). The exponent $1+\epsilon\varphi$ is the strategic complementarity familiar from Woodford (2003, ch. 3, eq. 1.35). It lowers the log-linear slope of the Phillips curve by the factor $1/(1+\epsilon\varphi)$.

2.4 Rotemberg benchmark

Under Rotemberg pricing, firms pay a quadratic adjustment cost $\frac{\psi}{2}(\Pi_t - 1)^2 Y_t$, so $C_t = [1 - \frac{\psi}{2}(\Pi_t - 1)^2]Y_t$ and $N_t = Y_t$. The symmetric pricing condition is

$$\psi(\Pi_t - 1)\Pi_t = (1-\epsilon) + \epsilon\, MC_t + \beta_t \left(\frac{C_{t+1}}{C_t}\right)^{-\sigma} \frac{Y_{t+1}}{Y_t}\, \psi(\Pi_{t+1} - 1)\Pi_{t+1}, \qquad MC_t = \chi Y_t^{\varphi} C_t^{\sigma}. \tag{9}$$

We set $\psi = (\epsilon - 1)\theta/[(1-\theta)(1-\beta\theta)]$, so that the log-linear Phillips curve coincides with that of the economy-wide Calvo model. This is the mapping in Eggertsson and Singh (2019, eq. 11) with their $\omega = 0$.

2.5 Shock and policy

For $0 \le t < T$ the discount factor is $\beta_t = 1/(1+\bar{r})$ with $\bar{r} < 0$, so that $r_t^n = \bar{r}$, and the nominal rate is held at zero. For $t \ge T$, $\beta_t = \beta$ and policy delivers strict price stability, $\Pi_t = 1$, which is the nonlinear counterpart of the terminal condition $\pi_T = x_T = 0$ used by Werning (2012). The trap therefore covers periods $0, \ldots, T-1$, so period 0 is the first trap period and period T is the first period after exit. The economy enters the trap from the zero-inflation steady state, $S_{-1} = 1$. It is convenient to define

$$\bar{x} \equiv \sigma^{-1} \log(1 + \bar{r}) < 0. \tag{10}$$

This is the output gap that the Euler equation (2) implies in the last trap period when inflation is zero and output is at its steady-state level from T on.

The log-linear benchmark is

$$x_t = x_{t+1} - \sigma^{-1}(i_t - \pi_{t+1} - r_t^n), \qquad \pi_t = \beta\pi_{t+1} + \kappa x_t, \tag{11}$$

where $\pi_t \equiv \log \Pi_t$, and with $\kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta}\zeta$ in the economy-wide case and $\kappa/(1+\epsilon\varphi)$ in the firm-specific case. Solved backward from $\pi_T = x_T = 0$, it has a unique bounded solution for every $\theta \in (0,1)$, and as $\theta \to 0$ inflation and output during the trap diverge to $-\infty$ (see Werning, 2012, Proposition 2, for the continuous-time version).

The existence of a unique bounded solution for every θ is specific to the deterministic model. When the trap instead ends with a constant probability each period, the log-linearized model has a locally unique bounded solution only if the shock is not too persistent relative to the

slope of the Phillips curve, which is condition C2 in Section 1.1 (Eggertsson and Singh, 2019, eq. 10). At the critical persistence the solution diverges, and beyond it the log-linearized model is indeterminate, so it too loses its unique equilibrium at a positive degree of stickiness.

3 Properties of Calvo Aggregation

We first show that strict price stability after the trap is feasible in the exact model, so that a failure of existence has to come from the trap periods. We then derive two bounds implied by the Calvo price index, one on the reset price and one on price dispersion.

Lemma 1 (Terminal regime)**.** *Let $\Pi_t = 1$ for all $t \geq T$. With firm-specific labor markets, this is an equilibrium if and only if $Y_t = Y$ for all $t \geq T$. With an economy-wide labor market, it is an equilibrium if and only if $MC_t = MC$ for all $t \geq T$, i.e.*

$$Y_t = Y\, S_t^{-\varphi/\zeta}, \qquad S_t = (1-\theta) + \theta S_{t-1}, \qquad t \geq T. \tag{12}$$

In both cases the implied nominal rate is strictly positive for $t \geq T$.

In the economy-wide labor market case, output stays slightly below steady state while price dispersion decays, with the gap $x_t = -(\varphi/\zeta)\log S_t$ being second-order in past inflation. By Lemma 1, strict price stability after the exit from the zero lower bound is feasible under both labor-market structures, and price dispersion alone does not invalidate the standard terminal condition. The results below keep strict price stability as the terminal regime.

Lemma 2 (Reset-price floor)**.** *For $\theta \in (0,1)$ and $\epsilon > 1$, any solution of Equation (5) with $\Pi_t \in (0,\infty)$ satisfies*

$$Z_t > \underline{Z}(\theta) \equiv (1-\theta)^{1/(\epsilon-1)}, \qquad \Pi_t < \theta^{-1/(\epsilon-1)}. \tag{13}$$

Moreover $\Pi_t \leq 1$ if and only if $Z_t \leq 1$, and $\Pi_t \to 0$ if and only if $Z_t \to \underline{Z}(\theta)$.

Note that $\underline{Z}(\theta) \to 1$ as $\theta \to 0$: when nearly all firms reset each period, the price level must be nearly equal to the reset price, however steep the deflation.

Lemma 3 (Dispersion under non-positive inflation)**.** *$S_t \geq 1$ for all t. Let $B(\theta) \equiv (1-\theta)^{-\epsilon/(\epsilon-1)}$. If $S_{t-1} \leq B(\theta)$ and $\Pi_t \leq 1$, then $S_t \leq B(\theta)$. Hence along any path starting at $S_{-1} = 1$ with $\Pi_t \leq 1$ for all t, $1 \leq S_t \leq B(\theta)$, and $B(\theta) \to 1$ as $\theta \to 0$.*

The lemma bounds dispersion in a deflationary episode: firms stuck at high prices sell little, and the reset-price floor in Lemma 2 caps the dispersion that resetting firms introduce. Dispersion under deflation is negligible near the flexible-price limit because $B(\theta) \to 1$ as $\theta \to 0$.

4 The Last Trap Period: Analytical Results

Consider period $T-1$, the last period of the trap, and suppose the terminal regime of Lemma 1 holds from T. With $\Pi_T = 1$ and $i_{T-1} = 0$, the Euler equation (2) gives $Y_{T-1} = (1+\bar{r})^{1/\sigma} Y_T$, independently of θ. Real marginal cost equals MC from T onward, so $\frac{\epsilon}{\epsilon-1} V_{1,T} = V_{2,T}$ (and likewise for $V^I_{1,T}$), and the reset condition at $T-1$ is a weighted average, given in Equation (15) below. To state this average, define

$$\eta(\theta) \equiv \frac{Y_{T-1}^{1-\sigma}}{Y_{T-1}^{1-\sigma} + \beta_{T-1}\theta V_{2,T}} \in (0,1). \tag{14}$$

The denominator of $\eta(\theta)$ equals $V_{2,T-1}$ because $\Pi_T = 1$. Hence $\eta(\theta)$ is the weight of period $T-1$ in the discounted sum that a resetting firm uses to value its sales. A price set at $T-1$ stays in place in each later period with probability θ, so later periods receive more weight when θ is larger. With firm-specific labor markets $\eta(\theta)$ is strictly decreasing in θ (see the proof of Proposition 1), and in both labor-market cases $\eta(\theta) \to 1$ as $\theta \to 0$ (Appendix A.7). If $\sigma = 1$, $\eta(\theta) = [1 + \beta_{T-1}\theta/(1-\beta\theta)]^{-1}$. The reset condition at $T-1$ is then (Appendix A.7)

$$\frac{\epsilon}{\epsilon - 1} \frac{V_{1,T-1}}{V_{2,T-1}} = \eta(\theta)\, m + \big(1 - \eta(\theta)\big), \tag{15}$$

where m is real marginal cost at $T-1$ relative to steady state. Relative marginal cost equals one from T on, and the reset condition averages m and this value of one with weights $\eta(\theta)$ and $1-\eta(\theta)$.

4.1 Firm-specific labor markets

Here $Y_T = Y$, so $x_{T-1} = \bar{x}$ and $m = e^{\zeta \bar{x}} < 1$. The next proposition shows that the reset-price floor alone rules out an equilibrium once prices are flexible enough. The proposition follows from combining Equations (8) and (15) with Lemma 2.

Proposition 1 (Existence boundary, firm-specific labor)**.** *An equilibrium in period $T-1$ exists*

if and only if

$$\eta(\theta)\left(1-e^{\zeta\bar{x}}\right) \;<\; 1-(1-\theta)^{(1+\epsilon\varphi)/(\epsilon-1)}, \tag{16}$$

where $\eta(\theta)$ is defined in Equation (14)*. There is a unique $\theta_I^* \in (0,1)$ such that Inequality* (16) *holds if and only if $\theta > \theta_I^*$, so θ_I^* is the existence boundary with firm-specific labor markets. As $\theta \downarrow \theta_I^*$, $\Pi_{T-1} \to 0$. If $T \geq 2$, then also $Y_{T-2} \to 0$. To first order in $|\bar{x}|$, the existence boundary satisfies*

$$\theta_I^* \approx \frac{(\epsilon-1)\,\zeta|\bar{x}|}{1+\epsilon\varphi}. \tag{17}$$

The equilibrium allocation in the firm-specific case is independent of price dispersion, as emphasized by Eggertsson and Singh (2019). The existence boundary nonetheless comes from aggregation, through the price index (5), which is independent of the labor-market structure. Equivalently, if all resetting firms chose $Z_{T-1} \leq \underline{Z}(\theta)$, the CES aggregate of their outputs would exceed the output level Y_{T-1} assumed in their demand curves.

4.2 Economy-wide labor market

Here $Y_T = Y S_T^{-\varphi/\zeta}$ with $S_T = 1-\theta+\theta S_{T-1}$, so real marginal cost at $T-1$ relative to steady state is

$$m = e^{\zeta\bar{x}}\left(\frac{S_{T-1}}{1-\theta+\theta S_{T-1}}\right)^{\varphi}. \tag{18}$$

Dispersion raises marginal cost at $T-1$ relative to T and pushes the reset price up. By Lemma 3 this effect vanishes near the flexible-price limit along a deflationary path. The next proposition shows that the reset-price floor again rules out a deflationary equilibrium once prices are flexible enough, and its proof uses the bound on dispersion in Lemma 3.

Proposition 2 (Existence boundary, economy-wide labor)**.** *An equilibrium in period $T-1$ with $\Pi_{T-1} \leq 1$ requires*

$$\eta(\theta)\,(1-m) \;<\; 1-(1-\theta)^{1/(\epsilon-1)}, \tag{19}$$

where $\eta(\theta)$ is defined in Equation (14) *and m in Equation* (18)*. If $S_{T-2} \leq B(\theta)$, which holds whenever inflation has been non-positive since the trap began, there exists $\underline{\theta} > 0$ such that no equilibrium with $\Pi_{T-1} \leq 1$ exists for $\theta < \underline{\theta}$. Let θ_C^* denote the existence boundary with an economy-wide labor market, defined as the value of θ at which the deflationary branch, followed downward from high θ, terminates. Then $\theta_C^* \geq \underline{\theta}$. If this branch terminates at the reset-price floor, as it does numerically, then $\Pi_{T-1} \to 0$ and $S_{T-1} \to (1-\theta)^{-1/(\epsilon-1)}$ as $\theta \downarrow \theta_C^*$, and to first*

order in $|\bar{x}|$ and θ the existence boundary satisfies

$$\theta_C^* \approx \frac{(\epsilon - 1)\,\zeta|\bar{x}|}{1 + \varphi}. \tag{20}$$

Proposition 2 is weaker than Proposition 1, because of difficulties in deriving the exact existence boundary in the economy-wide labor-market case. The threshold θ_C^* is the last existence boundary we can demonstrate via the numerical analysis.

At our baseline calibration ($\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$, $\bar{r} = -0.5\%$ per quarter), the first-order formulas give $\theta_C^* \approx 0.025$ and $\theta_I^* \approx 0.007$, and the exact numerical thresholds for $T = 1$ are 0.025 and 0.007 (Table 1). The economy-wide threshold is larger because the economy-wide model has a steeper Phillips curve: the same output gap calls for a larger reset-price cut, which the reset-price floor rules out at a higher value of θ.

Proposition 2 proves that some $\underline{\theta} > 0$ exists but does not give a value. Its proof bounds m from above. If $S_{T-2} \leq B(\theta)$ and $\Pi_{T-1} \leq 1$, Lemma 3 gives $S_{T-1} \leq B(\theta)$, and because $S/(1 - \theta + \theta S)$ is increasing in S, Equation (18) implies $m \leq \bar{m}(\theta)$, where

$$\bar{m}(\theta) \equiv e^{\zeta \bar{x}} \left(\frac{B(\theta)}{1 - \theta + \theta B(\theta)} \right)^{\varphi}. \tag{21}$$

Replacing m with $\bar{m}(\theta)$ in Inequality (19) gives a sufficient condition for non-existence. If

$$\eta(\theta) \left(1 - \bar{m}(\theta)\right) \;\geq\; 1 - (1 - \theta)^{1/(\epsilon - 1)}, \tag{22}$$

then Inequality (19) fails, and no equilibrium in period $T - 1$ with $\Pi_{T-1} \leq 1$ exists. In Inequality (22) the bound $\bar{m}(\theta)$ is explicit, but $\eta(\theta)$ in general depends on the path of output after T. With $\sigma = 1$, $\eta(\theta)$ in Equation (14) has a closed form, so Inequality (22) becomes an explicit condition on θ. The next proposition uses this condition to give an explicit $\underline{\theta}$ and its first-order approximation.

Proposition 3 (Explicit bound, economy-wide labor, $\sigma = 1$)**.** *Let $\sigma = 1$ and $S_{T-2} \leq B(\theta)$, and define*

$$G(\theta) \equiv \frac{1 - e^{\zeta \bar{x}} \left[B(\theta) / \left(1 - \theta + \theta B(\theta)\right) \right]^{\varphi}}{1 + \beta_{T-1} \theta / (1 - \beta \theta)} - \left[1 - (1 - \theta)^{1/(\epsilon - 1)}\right], \tag{23}$$

with $B(\theta)$ as in Lemma 3. The function $G(\theta)$ is the left side of Inequality (22) minus its right side, with $\eta(\theta)$ in closed form. If $G(\theta) \geq 0$, no equilibrium in period $T-1$ with $\Pi_{T-1} \leq 1$ exists.

The value

$$\underline{\theta} \equiv \inf\{\theta \in (0,1) : G(\theta) < 0\} \tag{24}$$

is a lower bound on the existence boundary θ_C^* of Proposition 2, with $0 < \underline{\theta} \leq \theta_C^*$. No equilibrium with $\Pi_{T-1} \leq 1$ exists for $\theta < \underline{\theta}$, and to first order in $|\bar{x}|$ and θ the lower bound satisfies

$$\underline{\theta} \approx \frac{(\epsilon - 1)\,\zeta|\bar{x}|}{1 + \epsilon\varphi}. \tag{25}$$

4.3 Rotemberg pricing

Under Rotemberg pricing, the terminal regime is simply $\Pi_t = 1$, $C_t = Y_t = Y$ for $t \geq T$. At $T-1$, $C_{T-1} = Ye^{\bar{x}}$ and Equation (9) becomes $F(\Pi) \equiv \psi(\Pi - 1)\Pi - (\epsilon - 1)(m_R(\Pi) - 1) = 0$ with

$$m_R(\Pi) = e^{\zeta\bar{x}}\left[1 - \tfrac{\psi}{2}(\Pi - 1)^2\right]^{-\varphi}, \tag{26}$$

since the adjustment cost raises output relative to consumption, and higher output raises labor and marginal cost.

The Rotemberg model has no reset-price floor. The next proposition shows that its deflationary equilibrium also disappears for small θ, through the effect of adjustment costs on marginal cost.

Proposition 4 (Existence boundary, Rotemberg)**.** *Let $\psi < 2$. (a) If $(\epsilon - 1)\big[1 - e^{\zeta\bar{x}}(1 - \psi/2)^{-\varphi}\big] > \psi/4$, no equilibrium with $\Pi_{T-1} \in (0,1]$ exists. (b) If $e^{\zeta\bar{x}}(1 - \psi/2)^{-\varphi} > 1$ or $\psi/4 > (\epsilon - 1)(1 - e^{\zeta\bar{x}})$, an equilibrium with $\Pi_{T-1} \in (0,1)$ exists. Let θ_R^* denote the existence boundary of the Rotemberg model, defined as the infimum of the values of θ for which an equilibrium with $\Pi_{T-1} \in (0,1)$ exists. To first order in $|\bar{x}|$, with $\psi \approx (\epsilon - 1)\theta$ for small θ and ϵ large enough that $2(\epsilon - 1)\varphi \gg 1$, the existence boundary satisfies*

$$\theta_R^* \approx \frac{2\zeta|\bar{x}|}{(\epsilon - 1)\varphi}. \tag{27}$$

The Calvo and Rotemberg models share a log-linear approximation but reach their existence boundaries through different mechanisms. In the Calvo model the existence boundary arises from aggregation through the reset-price floor in Lemma 2, and the distance $1 - \underline{Z}(\theta)$ between one and the reset-price floor is approximately $\theta/(\epsilon - 1)$. In the Rotemberg model the existence boundary arises from the resource cost of deflation, and this cost raises marginal cost by roughly

$\varphi\psi/2 \approx \varphi(\epsilon-1)\theta/2$. For conventional elasticities of substitution this cost effect is much larger than $1-\underline{Z}(\theta)$, so in a one-period trap the Rotemberg model accommodates far more flexibility before its deflationary equilibrium disappears.

Proposition 4 assumes $\psi < 2$, which holds near the thresholds of a one-period trap. In longer traps the relevant values of θ are larger, ψ exceeds 2, and the resource constraint becomes the binding limit, as Section 5 shows.

5 Multi-Period Traps

We solve the exact perfect-foresight model for traps of $T \in \{1, 2, 4, 8\}$ quarters and $\bar{r} \in \{-0.25\%, -0.5\%, -1\%\}$ per quarter, with the baseline calibration $\sigma = \varphi = 1$, $\epsilon = 6$ and $\beta = 0.99$. The firm-specific Calvo model and the Rotemberg model are solved by exact backward recursion. The economy-wide Calvo model, in which dispersion is a backward-looking state, is solved by a Newton method with continuation in θ, imposing the reset-price floor of Lemma 2 by construction. In each model we follow the deflationary branch, the solution connected to the log-linear one at high θ. Appendix C describes the methods and their computational validation.

The numerical results involve no approximation of the model. For each value of θ we solve the nonlinear equilibrium conditions without linearization, up to the solver tolerances in Appendix C, and we locate each threshold by searching over θ. We call the results numerical only because we computationally and iteratively solve over different values of θ to answer questions such as where the existence boundary lies, so that we can cross-validate the analytical results and extend them to longer traps.

5.1 Trap length

One property of longer traps holds analytically in two of the three models. The last T periods of a trap of length $T+1$ face the same shock and the same terminal regime as a trap of length T. Whenever no state variable links the first trap period to later periods, a longer trap therefore cannot widen the range of θ for which an equilibrium exists.

Proposition 5 (Monotonicity in trap length)**.** *Consider either firm-specific labor markets or Rotemberg pricing. For $T \geq 1$, let Θ_T denote the set of $\theta \in (0,1)$ for which an equilibrium with a trap of length T exists, and let $\theta^*(T) \equiv \inf \Theta_T$, with $\theta^*(T) = 1$ if Θ_T is empty. Then $\Theta_{T+1} \subseteq \Theta_T$ for every $T \geq 1$, so the existence threshold $\theta^*(T)$ is nondecreasing in T.*

Table 1: θ existence thresholds and the Rotemberg 50% cost threshold. For C-E, C-I and R, the deflationary equilibrium exists if and only if $\theta > \theta^*$. For R50, Rotemberg adjustment costs stay below half of output at every date if and only if $\theta > \theta^*$.

	$\bar{r} = -0.25\%$				$\bar{r} = -0.5\%$				$\bar{r} = -1\%$			
T	C-E	C-I	R	R50	C-E	C-I	R	R50	C-E	C-I	R	R50
1	0.012	0.004	0.002	0.002	0.025	0.007	0.004	0.004	0.049	0.014	0.008	0.008
2	0.114	0.036	0.055	0.055	0.163	0.053	0.073	0.073	0.231	0.080	0.097	0.097
4	0.353	0.147	0.207	0.271	0.425	0.188	0.221	0.323	0.511	0.243	0.231	0.384
8	0.622	0.360	0.235	0.571	0.686	0.419	0.235	0.616	0.755	0.491	0.235	0.666

Notes: C-E: Calvo, economy-wide labor market. C-I: Calvo, firm-specific labor markets. R: Rotemberg with log-linear slope matched to C-E, existence threshold. R50: smallest θ at which Rotemberg adjustment costs stay below 50% of output at every date. $\bar{r}$ is per quarter and T is in quarters. The deterministic log-linear model has a solution for every $\theta \in (0, 1)$. The trap length T and the natural rate $\bar{r}$ are as shown in each row, and all other parameters are at the baseline calibration $\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$.

The argument does not apply with an economy-wide labor market. Marginal cost depends on the price-dispersion state variable through Equation (4). We instead resort to the numerical analysis to confirm monotonicity in T and leave the analytic analysis to future work.

5.2 Rotemberg pricing and the resource constraint

Adjustment costs in the Rotemberg model are paid in output, so $C_t = k_t Y_t$ with $k_t \equiv 1 - \frac{\psi}{2}(\Pi_t - 1)^2$. When $\psi > 2$, Equation (9) at each date has a root $\Pi_t > 1 - \sqrt{2/\psi}$, so the deflationary equilibrium exists however long the trap, although adjustment costs can absorb almost all output. Existence is lost only once ψ falls below 2.

Eggertsson and Singh (2019) argue that solutions of this kind are not economically meaningful. In their calibration of a Great Depression, a deflation of 10% would require price-adjustment costs above total output, and they conclude that the Rotemberg model with quadratic resource costs is not suited to the study of extreme events. We therefore report two Rotemberg thresholds: the existence threshold, and the smallest θ at which adjustment costs stay below half of output at every date. Since the 50% threshold is not an existence condition, the Rotemberg equilibrium may still exist below the threshold, but we consider such an equilibrium not economically meaningful. We also note that the 50% threshold provides the fairer comparison with Calvo pricing as well.

Table 1 reports the thresholds. The Calvo thresholds rise steeply with the length of the trap, because the reset price at each date depends on deflation expected at all later dates of the trap, so the reset-price floor binds in the first trap period at much higher values of θ. Proposition 5 shows that the existence thresholds of the firm-specific Calvo and Rotemberg models cannot

Table 2: First trap period, $T = 8$, $\bar{r} = -0.5\%$: $100 \log \Pi_0$ and $100\, x_0$.

	Log-lin. (E)		Calvo-E		Rotemberg		Log-lin. (I)		Calvo-I	
θ	π_0	x_0	π_0	x_0	π_0	x_0	π_0	x_0	π_0	x_0
0.90	−0.5	−5.1	−0.5	−5.1	−0.5	−5.1	−0.1	−4.2	−0.1	−4.1
0.85	−1.4	−6.7	−1.7	−7.0	−1.3	−6.7	−0.1	−4.3	−0.1	−4.3
0.80	−3.2	−9.9	−4.8	−10.3	−2.8	−9.5	−0.3	−4.7	−0.3	−4.7
0.75	−7.4	−15.7	−12.9	−14.9	−5.6	−14.5	−0.5	−5.1	−0.5	−5.1
0.70	−17.2	−27.1	−41.1	−21.2	−10.5	−22.9	−0.8	−5.8	−0.9	−5.8

Notes: Period 0 is the first trap period, and $\pi_0 = 100 \log \Pi_0$ and x_0 (in percent) are inflation and the output gap in that period. (E) and (I) denote the Phillips-curve slope of the economy-wide and firm-specific cases. For Rotemberg, x_0 is the consumption gap. Price dispersion in Calvo-E at $t = 0$ is $S_0 = 1.001, 1.004, 1.019, 1.070, 1.196$ for the five values of θ. The trap lasts $T = 8$ quarters with $\bar{r} = -0.5\%$ per quarter, and all other parameters are at the baseline calibration $\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$.

fall as T rises, and Table 1 shows that the existence thresholds rise in all three models. For $T = 8$ and $\bar{r} = -1\%$ (about -4% annualized), the economy-wide Calvo threshold is 0.755, so the deflationary equilibrium does not exist at the conventional calibration $\theta = 0.75$. The firm-specific threshold lies below the economy-wide one because the firm-specific model has a flatter Phillips curve.

In long traps the Rotemberg equilibrium needs adjustment costs above half of output at somewhat lower stickiness than the point where the economy-wide Calvo equilibrium ceases to exist: 0.616 against 0.686 at $T = 8$ and $\bar{r} = -0.5\%$.

Table 2 compares the first trap period across models. With firm-specific labor markets, the nonlinear and log-linear solutions are nearly identical at the values of θ considered, which confirms the finding of Eggertsson and Singh (2019) away from the existence boundary. With an economy-wide labor market, the nonlinear model delivers much more deflation than its log-linear approximation as θ falls (41 against 17 log points at $\theta = 0.7$), while the output gap at $t = 0$ is somewhat smaller because it depends on inflation at later dates. The Rotemberg model with the same log-linear slope produces less deflation than either, as in Miao and Ngo (2021). Nonlinear Calvo and Rotemberg pricing thus diverge well before either reaches a threshold.

Table 3 shows how each model branch ends as θ is lowered. In both Calvo models, inflation in the first trap period diverges to $-\infty$ as $\theta \downarrow \theta^*$, while the log-linearized model with the same Phillips-curve slope gives finite inflation at the same values of θ. Near the existence boundary the exact and log-linearized solutions therefore diverge significantly even with firm-specific labor markets, so the accuracy of the log-linear approximation that Eggertsson and Singh (2019) find holds only away from the existence boundary. Output in the first trap period stays finite,

Table 3: Approach to the existence boundary, $T = 8$, $\bar{r} = -0.5\%$: exact and log-linearized first-period outcomes and, for Rotemberg, the smallest $\log k_t$ along the path.

	Calvo-E ($\theta^* = 0.686$)			Calvo-I ($\theta^* = 0.419$)			Rotemberg ($\theta^* = 0.235$)		
$\theta - \theta^*$	π_0	x_0	π_0^{lin}	π_0	x_0	π_0^{lin}	π_0	x_0	$\min_t \log k_t$
2×10^{-2}	-33.9	-20.3	-15.6	-26.7	-24.8	-10.5	-273.4	-1074.8	-21.5
10^{-3}	-96.6	-23.4	-21.6	-82.3	-30.7	-13.0	-567.9	-2052.8	-41.1
10^{-5}	-192.0	-23.6	-21.9	-174.2	-31.1	-13.2	-1028.1	-3216.4	-64.3

Notes: Period 0 is the first trap period, and $\pi_0 = 100 \log \Pi_0$ and x_0 (in percent) are inflation and the output gap in that period in the exact model. π_0^{lin} is first-period inflation in the log-linearized model with the same Phillips-curve slope at the same θ. For Rotemberg, x_0 is the consumption gap. The trap lasts $T = 8$ quarters with $\bar{r} = -0.5\%$ per quarter, and all other parameters are at the baseline calibration $\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$.

Table 4: Time aggregation: quarterly-equivalent thresholds $(\theta^*_{\text{sub}})^q$ for $T = 8$ quarters, $\bar{r} = -0.5\%$ per quarter.

Sub-periods per quarter, q	1	3	6	13
Calvo-E	0.686	0.661	0.654	n.a.
Calvo-I	0.419	0.350	0.329	0.316
Rotemberg, existence	0.235	0.013	2×10^{-4}	1×10^{-8}
Rotemberg, costs $< 50\%$	0.616	0.600	0.594	0.590

Notes: The model is re-solved with q sub-periods per quarter, $\beta^{1/q}$ and $(1+\bar{r})^{1/q}$. The threshold on the sub-period Calvo parameter is converted to a quarterly probability of not resetting. Rotemberg uses the matched slope at each frequency. n.a.: not computed (the economy-wide solver has $8q$ unknowns). The trap lasts $T = 8$ quarters with $\bar{r} = -0.5\%$ per quarter, and all other parameters are at the baseline calibration $\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$, with β converted to the sub-period as above.

because it depends on inflation at later dates. In the Rotemberg model deflation also diverges near its existence boundary, but by then consumption is a vanishing fraction of output. In all three models the slump becomes deeper as θ approaches the existence boundary.

The reset-price floor of Lemma 2 is derived in discrete time, so one may ask whether the existence boundary is an artifact of discretization and period length. Table 4 re-solves the model at monthly, twice-monthly and weekly frequency, holding the trap length and the quarterly natural rate fixed. The Calvo thresholds decline modestly and appear to converge as the period shortens. The Rotemberg existence threshold instead falls toward zero, because the matched adjustment cost $\psi = (\epsilon - 1)\theta_{\text{sub}}/[(1 - \theta_{\text{sub}})(1 - \beta^{1/q}\theta_{\text{sub}})]$ depends on the Calvo parameter θ_{sub} of one sub-period. With q sub-periods per quarter, a quarterly probability θ of not resetting corresponds to $\theta_{\text{sub}} = \theta^{1/q}$, which rises toward one as q increases. A given degree of quarterly stickiness therefore implies a larger ψ when the quarter has more sub-periods, and the condition $\psi > 2$ holds at ever lower quarterly stickiness. The 50% cost threshold changes little across frequencies. A formal continuous-time analysis is left for future work.

Table 5 compares the first-order formulas of Propositions 1, 2 and 4 with the exact thresholds

Table 5: Last trap period ($T = 1$): first-order formulas against exact thresholds.

	Calvo-E		Calvo-I		Rotemberg	
$\bar{r}$	formula	exact	formula	exact	formula	exact
-0.25%	0.0125	0.0124	0.0036	0.0036	0.0020	0.0020
-0.5%	0.0251	0.0247	0.0072	0.0071	0.0040	0.0040
-1%	0.0503	0.0489	0.0144	0.0141	0.0080	0.0078

Notes: The trap lasts $T = 1$ quarter, $\bar{r}$ is as shown in each row, and all other parameters are at the baseline calibration $\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$.

Table 6: Sensitivity to the calibration: θ existence thresholds and the Rotemberg 50% cost threshold for $T = 8$, $\bar{r} = -0.5\%$.

	Calvo-E	Calvo-I	Rotemberg, existence	Rotemberg, costs $< 50\%$
Baseline	0.686	0.419	0.235	0.616
$\epsilon = 4$	0.638	0.443	0.315	0.621
$\epsilon = 11$	0.755	0.371	0.146	0.600
$\varphi = 0.5$	0.678	0.480	0.235	0.601
$\varphi = 2$	0.696	0.367	0.235	0.632
$\sigma = 2$	0.641	0.356	0.235	0.547

Notes: One parameter at a time is changed from the baseline $\sigma = \varphi = 1$, $\epsilon = 6$, $\beta = 0.99$, with a trap of $T = 8$ quarters and $\bar{r} = -0.5\%$ per quarter.

for a one-period trap. The formulas are within about 3% of the exact thresholds at every shock size, with the error growing in $|\bar{r}|$ as expected.

Table 6 varies the calibration one parameter at a time for the two-year trap. The existence thresholds change with the calibration, but in every case the deflationary equilibrium ceases to exist at a strictly positive degree of price stickiness.

The thresholds do not depend on the initial dispersion S_{-1}. Steady-state dispersion under 2% annual trend inflation at $\theta = 0.75$ is about 1.001, and raising S_{-1} from 1 to 1.005 leaves the economy-wide thresholds for $T = 1$ and $T = 8$ unchanged to more than ten digits. This invariance is expected because inherited dispersion enters S_0 only through the term $\theta\Pi_0^{\epsilon}S_{-1}$ in Equation (6), and this term vanishes as $\Pi_0 \to 0$ at the existence boundary.

6 Discussion: Equilibrium Selection

Our results take the terminal condition $\Pi_t = 1$ for $t \geq T$ as given, as in Werning (2012). The non-existence of a deflationary equilibrium only suggests that the chosen equilibrium selection, in particular the terminal condition $\Pi_t = 1$ for $t \geq T$, does not work, and we may have to go for different equilibrium selections, as Cochrane (2017) suggests. Other selections still yield

equilibria. For instance, at $\theta = 0$ output equals its natural level in every period, and the Euler equation at $T-1$ requires $\Pi_T = 1/(1+\bar{r}) > 1$: a flexible-price equilibrium exists only if the price level is allowed to jump up at exit. Werning (2012) makes the same point for the log-linearized model: if the equilibria for finite κ are required to have nonpositive inflation and $\pi_T = 0$, no equilibrium exists at $\kappa = \infty$. He attributes the discontinuity in outcomes to the monetary policy assumed at exit, which requires zero inflation at T, rather than to the private-sector equilibrium conditions.

7 Conclusion

The exact Calvo price index bounds the reset price below by $(1-\theta)^{1/(\epsilon-1)}$ times the price level. Under the standard terminal condition, this reset-price floor makes the deflationary zero lower bound equilibrium cease to exist at a strictly positive Calvo parameter, which we characterize analytically for the last trap period and by numerical solution of the exact model for longer traps. With firm-specific labor markets and under Rotemberg pricing, a longer trap never lowers the existence threshold. Near the existence boundary, deflation grows without bound. The paradox of flexibility therefore remains in the exact model, where the unbounded slump occurs at a strictly positive degree of price stickiness. Near the existence boundary the exact and log-linearized solutions diverge significantly even with firm-specific labor markets, so the accuracy of the log-linear approximation found by Eggertsson and Singh (2019) holds only away from the existence boundary. The existence boundary exists under both economy-wide and firm-specific labor markets, and under a matched log-linear slope it is reached at much higher stickiness than in the Rotemberg model. For long traps it can lie above conventional calibrations of price stickiness. The non-existence of the deflationary equilibrium suggests that the terminal condition $\Pi_t = 1$ for $t \geq T$ does not work once prices are flexible enough, so that other equilibrium selections may be needed.

Data availability

The codes are provided at `https://mkimacad.github.io/nk_zlb/codes/code.zip`. Given its length, the codebase is not reproduced in this manuscript. The codes are also provided as part of the submission package.

Declaration of interests and ethical statements

The authors have no funding sources or conflicts of interest to declare. No experiment that requires ethical statements was performed.

References

Ascari, Guido, and Argia M. Sbordone. 2014. "The Macroeconomics of Trend Inflation." *Journal of Economic Literature*, 52(3): 679–739.

Ascari, Guido, and Lorenza Rossi. 2012. "Trend Inflation and Firms Price-Setting: Rotemberg Versus Calvo." *Economic Journal*, 122(563): 1115–1141.

Benhabib, Jess, Stephanie Schmitt-Grohé, and Martín Uribe. 2001. "The Perils of Taylor Rules." *Journal of Economic Theory*, 96(1–2): 40–69.

Boneva, Lena Mareen, R. Anton Braun, and Yuichiro Waki. 2016. "Some Unpleasant Properties of Loglinearized Solutions When the Nominal Rate Is Zero." *Journal of Monetary Economics*, 84: 216–232.

Cochrane, John H. 2017. "The New-Keynesian Liquidity Trap." *Journal of Monetary Economics*, 92: 47–63.

Eggertsson, Gauti B. 2011. "What Fiscal Policy Is Effective at Zero Interest Rates?" In *NBER Macroeconomics Annual 2010*. Vol. 25, 59–112. University of Chicago Press.

Eggertsson, Gauti B., and Michael Woodford. 2003. "The Zero Bound on Interest Rates and Optimal Monetary Policy." *Brookings Papers on Economic Activity*, 2003(1): 139–211.

Eggertsson, Gauti B., and Paul Krugman. 2012. "Debt, Deleveraging, and the Liquidity Trap: A Fisher-Minsky-Koo Approach." *Quarterly Journal of Economics*, 127(3): 1469–1513.

Eggertsson, Gauti B., and Sanjay R. Singh. 2019. "Log-linear Approximation versus an Exact Solution at the ZLB in the New Keynesian Model." *Journal of Economic Dynamics and Control*, 105: 21–43.

Fernández-Villaverde, Jesús, Grey Gordon, Pablo Guerrón-Quintana, and Juan F. Rubio-Ramírez. 2015. "Nonlinear Adventures at the Zero Lower Bound." *Journal of Economic Dynamics and Control*, 57: 182–204.

Galí, Jordi. 2008. *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework.* Princeton, NJ:Princeton University Press.

Gavin, William T., Benjamin D. Keen, Alexander W. Richter, and Nathaniel A. Throckmorton. 2015. "The Zero Lower Bound, the Dual Mandate, and Unconventional Dynamics." *Journal of Economic Dynamics and Control*, 55: 14–38.

Krugman, Paul R. 1998. "It's Baaack: Japan's Slump and the Return of the Liquidity Trap." *Brookings Papers on Economic Activity*, 1998(2): 137–205.

Miao, Jianjun, and Phuong V. Ngo. 2021. "Does Calvo Meet Rotemberg at the Zero Lower Bound?" *Macroeconomic Dynamics*, 25(4): 1090–1111.

Werning, Iván. 2012. "Managing a Liquidity Trap: Monetary and Fiscal Policy." Massachusetts Institute of Technology Working Paper. Revised version of NBER Working Paper 17344 (2011).

Woodford, Michael. 2003. *Interest and Prices: Foundations of a Theory of Monetary Policy.* Princeton, NJ:Princeton University Press.

Yun, Tack. 1996. "Nominal Price Rigidity, Money Supply Endogeneity, and Business Cycles." *Journal of Monetary Economics*, 37(2–3): 345–370.

A Derivations

A.1 Household

With Lagrange multiplier Λ_t on the budget constraint $P_t C_t + D_t = (1 + i_{t-1}) D_{t-1} + W_t N_t + \Upsilon_t$, where D_t denotes bond holdings, and discounting $\prod_{j<t} \beta_j$, the first-order conditions are $C_t^{-\sigma} = \Lambda_t P_t$, $\Lambda_t = \beta_t (1 + i_t) \Lambda_{t+1}$ and $\chi N_t^{\varphi} = \Lambda_t W_t$, which give the Euler equation (2) and $W_t / P_t = \chi N_t^{\varphi} C_t^{\sigma}$. For the cross-check with Eggertsson and Singh (2019): our σ corresponds to $1/\tilde{\sigma}$ there, our φ to ω, our ϵ to their θ, our Calvo parameter θ to their α, and our χ to λ. Our discount-factor shock corresponds to their preference shock ξ_t, and we set their $G_t = 0$ and $\tau_t^w = 0$.

A.2 Price index and demand

Minimizing $\int_0^1 P_{i,t} C_{i,t} di$ subject to $C_t = (\int_0^1 C_{i,t}^{(\epsilon-1)/\epsilon} di)^{\epsilon/(\epsilon-1)}$ gives $C_{i,t} = (P_{i,t}/P_t)^{-\epsilon} C_t$ with $P_t = (\int_0^1 P_{i,t}^{1-\epsilon} di)^{1/(1-\epsilon)}$. The same expressions follow from a competitive final-good producer, as shown on p. 42 of Galí (2008).

A.3 Calvo recursions

Non-resetting firms have the same price distribution as all firms in $t-1$, because resetting firms are drawn independently of their prices. Splitting the integrals,

$$
\begin{aligned}
P_t^{1-\epsilon} &= (1-\theta)(P_t^*)^{1-\epsilon} + \theta P_{t-1}^{1-\epsilon}, \\
S_t &= (1-\theta) \left(\frac{P_t^*}{P_t} \right)^{-\epsilon} + \theta \left(\frac{P_{t-1}}{P_t} \right)^{-\epsilon} \int_0^1 \left(\frac{P_{i,t-1}}{P_{t-1}} \right)^{-\epsilon} di,
\end{aligned}
$$

which give Equations (5) and (6). With an economy-wide labor market, $N_t = \int_0^1 Y_{i,t} di = S_t Y_t$, so $W_t / P_t = \chi (S_t Y_t)^{\varphi} Y_t^{\sigma}$, which gives Equation (4).

A.4 Reset price, economy-wide labor market

A resetting firm maximizes

$$
\sum_{h \geq 0} \theta^h Q_{t,t+h} \left[P_t^* Y_{t+h|t} - P_{t+h} MC_{t+h} Y_{t+h|t} \right]
$$

with $Y_{t+h|t} = (P_t^*/P_{t+h})^{-\epsilon} Y_{t+h}$ and $Q_{t,t+h} = \big(\prod_{j=0}^{h-1} \beta_{t+j}\big)(Y_{t+h}/Y_t)^{-\sigma} P_t/P_{t+h}$. The first-order condition,

$$P_t^* \sum_{h\geq 0} \theta^h Q_{t,t+h} P_{t+h}^{\epsilon} Y_{t+h} = \frac{\epsilon}{\epsilon - 1} \sum_{h\geq 0} \theta^h Q_{t,t+h} P_{t+h}^{1+\epsilon} MC_{t+h} Y_{t+h},$$

divided by P_t, yields Equation (7). See pages 44 and 45 of Galí (2008), and Ascari and Sbordone (2014, eqs. 26 to 28) for the recursive form. With firm-specific labor markets, $\frac{\epsilon}{\epsilon-1} V_{1,t}^I$ and $V_{2,t}$ below correspond to K_t and F_t in Eggertsson and Singh (2019), whose reset-price condition is $P_t^*/P_t = (K_t/F_t)^{1/(1+\omega\theta)}$ in their notation.

A.5 Reset price, firm-specific labor markets

A firm charging P_t^* at $t+h$ produces $Y_{t+h|t} = (Z_t/\Pi_{t,t+h})^{-\epsilon} Y_{t+h}$, where $\Pi_{t,t+h} \equiv P_{t+h}/P_t$, and pays the wage of its firm-specific labor market, so its real marginal cost is

$$\chi Y_{t+h|t}^{\varphi} C_{t+h}^{\sigma} = MC_{t+h}^{s} \left(\frac{Z_t}{\Pi_{t,t+h}} \right)^{-\epsilon\varphi}.$$

Substituting into the first-order condition above and collecting powers of Z_t gives

$$Z_t^{1+\epsilon\varphi} = \frac{\epsilon}{\epsilon - 1} \frac{\sum_h (\beta\theta)_t^h \, \Pi_{t,t+h}^{\epsilon(1+\varphi)} Y_{t+h}^{1-\sigma} MC_{t+h}^{s}}{\sum_h (\beta\theta)_t^h \, \Pi_{t,t+h}^{\epsilon-1} Y_{t+h}^{1-\sigma}},$$

where $(\beta\theta)_t^h \equiv \theta^h \prod_{j=0}^{h-1} \beta_{t+j}$. This is Equation (8) in recursive form.

A.6 Rotemberg

With adjustment cost $\frac{\psi}{2}(P_{i,t}/P_{i,t-1} - 1)^2 Y_t$ in units of the aggregate good, the symmetric first-order condition is Equation (9), and $C_t = [1 - \frac{\psi}{2}(\Pi_t - 1)^2] Y_t$. Log-linearizing around the zero-inflation steady state gives the slope $(\epsilon - 1)\zeta/\psi$, which equals the Calvo slope when $\psi = (\epsilon - 1)\theta/[(1-\theta)(1-\beta\theta)]$.

A.7 The reset condition at $T-1$

From T on, $\Pi_t = 1$ and real marginal cost equals MC (Lemma 1), so $V_{1,T} = MC\, V_{2,T}$ and $V_{1,T}^I = MC\, V_{2,T}$, where $V_{2,T} = \sum_{h\geq 0} (\beta\theta)^h Y_{T+h}^{1-\sigma}$. Since $\Pi_T = 1$,

$$\frac{\epsilon}{\epsilon - 1} \frac{V_{1,T-1}}{V_{2,T-1}} = \frac{Y_{T-1}^{1-\sigma} \frac{MC_{T-1}}{MC} + \beta_{T-1}\theta V_{2,T}}{Y_{T-1}^{1-\sigma} + \beta_{T-1}\theta V_{2,T}} = \eta(\theta)\, m + \big(1 - \eta(\theta)\big),$$

with $\eta(\theta)$ as in Equation (14), which is Equation (15). The same holds for V^I. In the economy-wide case, $Y_{T-1} = (1+\bar{r})^{1/\sigma} Y_T$ and $Y_T = Y S_T^{-\varphi/\zeta}$ give $(Y_{T-1}/Y)^\zeta = e^{\zeta\bar{x}} S_T^{-\varphi}$, and Equation (18) follows from Equation (4). For $t \geq T$, $Y_t \in [YB(\theta)^{-\varphi/\zeta}, Y]$ along the paths considered, so $V_{2,T}$ is bounded and $\eta(\theta) \to 1$ as $\theta \to 0$.

B Proofs

Proof of Lemma 1. By Equation (5), $\Pi_t = 1$ if and only if $Z_t = 1$. With $\Pi_{t+1} = 1$ for all $t \geq T$, $Z_t = 1$ for all $t \geq T$ requires $V_{1,t} = MC\, V_{2,t}$ for all $t \geq T$ (economy-wide case). Subtracting MC times the V_2 recursion from the V_1 recursion gives $Y_t^{1-\sigma}(MC_t - MC) + \beta\theta(V_{1,t+1} - MC\, V_{2,t+1}) = 0$, so $MC_t = MC$ for all $t \geq T$. Conversely, $MC_t = MC$ for all $t \geq T$ implies $V_{1,t} = MC\, V_{2,t}$ and hence $Z_t = 1$. By Equation (4), $MC_t = MC$ is equivalent to $Y_t = Y S_t^{-\varphi/\zeta}$, and Equation (6) with $Z_t = \Pi_t = 1$ gives $S_t = 1 - \theta + \theta S_{t-1}$. The firm-specific case is identical with $MC_t^s = \chi Y_t^\zeta$, which equals MC if and only if $Y_t = Y$. For the nominal rate, Equation (2) with $\Pi_{t+1} = 1$ and $\beta_t = \beta$ gives $1 + i_t = \beta^{-1}(Y_{t+1}/Y_t)^\sigma$. Y_t is non-decreasing because S_t is non-increasing for $t \geq T$, so $1 + i_t \geq \beta^{-1} > 1$. □

Proof of Lemma 2. From Equation (5), $(1-\theta)Z_t^{1-\epsilon} = 1 - \theta\Pi_t^{\epsilon-1} < 1$, so $Z_t^{1-\epsilon} < (1-\theta)^{-1}$ and, since $1 - \epsilon < 0$, $Z_t > (1-\theta)^{1/(\epsilon-1)}$. Likewise $\theta\Pi_t^{\epsilon-1} < 1$. The map $Z \mapsto \Pi = [(1 - (1-\theta)Z^{1-\epsilon})/\theta]^{1/(\epsilon-1)}$ is continuous and strictly increasing on $(\underline{Z}, \infty)$, with $\Pi = 1$ at $Z = 1$ and $\Pi \to 0$ as $Z \downarrow \underline{Z}$, which gives the remaining statements. □

Proof of Lemma 3. Let $y_i = (P_{i,t}/P_t)^{1-\epsilon}$, so $\int_0^1 y_i di = 1$ by the definition of P_t. Then $S_t = \int_0^1 y_i^{\epsilon/(\epsilon-1)} di \geq (\int_0^1 y_i di)^{\epsilon/(\epsilon-1)} = 1$ by Jensen's inequality. For the upper bound, Lemma 2 gives $Z_t^{-\epsilon} < (1-\theta)^{-\epsilon/(\epsilon-1)}$, so with $\Pi_t \leq 1$ and $S_{t-1} \leq B$,

$$S_t \leq (1-\theta)(1-\theta)^{-\epsilon/(\epsilon-1)} + \theta B = (1-\theta)^{-\epsilon/(\epsilon-1)}\big[(1-\theta) + \theta\big] = B. \qquad \square$$

Proof of Proposition 1. By Equations (8) and (15), $Z_{T-1}^{1+\epsilon\varphi} = 1 - \eta(\theta)(1 - e^{\zeta\bar{x}}) \in (0,1)$, so $Z_{T-1} < 1$. By Lemma 2, a solution $\Pi_{T-1} \in (0,1)$ of Equation (5) exists if and only if $Z_{T-1} > (1-\theta)^{1/(\epsilon-1)}$, which is Inequality (16). With firm-specific labor markets $Y_t = Y$ for $t \geq T$, so $V_{2,T} = Y^{1-\sigma}/(1-\beta\theta)$, and Y_{T-1} does not depend on θ. Hence, by Equation (14), $\eta(\theta) = [1 + \beta_{T-1}\theta\,(Y/Y_{T-1})^{1-\sigma}/(1-\beta\theta)]^{-1}$ is strictly decreasing in θ, and the left side of Inequality (16) is strictly decreasing, positive at $\theta \to 0$ (where it tends to $1 - e^{\zeta\bar{x}} > 0$) and below one. The

right side is strictly increasing from 0 to 1. So the two sides have a unique crossing point $\theta_I^* \in (0,1)$. At θ_I^*, $Z_{T-1} = \underline{Z}$, and $\Pi_{T-1} \to 0$ by Lemma 2. If $T \geq 2$, Equation (2) gives $Y_{T-2}^{-\sigma} = \beta_{T-2} Y_{T-1}^{-\sigma} / \Pi_{T-1} \to \infty$. For the approximation, $\eta(\theta) = 1 - O(\theta)$ by Equation (14), $1 - e^{\zeta \bar{x}} = \zeta|\bar{x}| + O(\bar{x}^2)$ and $1 - (1-\theta)^{(1+\epsilon\varphi)/(\epsilon-1)} = \frac{1+\epsilon\varphi}{\epsilon-1}\theta + O(\theta^2)$. □

Proof of Proposition 2. By Equations (7) and (15), $Z_{T-1} = 1 - \eta(\theta)(1-m)$. If $\Pi_{T-1} \leq 1$ then, by Lemma 2, $Z_{T-1} > (1-\theta)^{1/(\epsilon-1)}$, which is Inequality (19). If $S_{T-2} \leq B(\theta)$ and $\Pi_{T-1} \leq 1$, Lemma 3 gives $S_{T-1} \leq B(\theta)$, and since $S \mapsto S/(1-\theta+\theta S)$ is increasing, $m \leq \bar{m}(\theta)$ with $\bar{m}(\theta)$ as in Equation (21), and Inequality (22) implies that Inequality (19) fails. As $\theta \to 0$, $B(\theta) \to 1$, $\bar{m}(\theta) \to e^{\zeta \bar{x}} < 1$ and $\eta(\theta) \to 1$ (Appendix A.7), so the left side of Inequality (22) tends to $1 - e^{\zeta \bar{x}} > 0$ while its right side tends to zero. Hence Inequality (22) holds, and Inequality (19) fails, for all sufficiently small θ. The deflationary branch has solutions at values of θ just above θ_C^*, and no solution with $\Pi_{T-1} \leq 1$ exists below $\underline{\theta}$, so $\theta_C^* \geq \underline{\theta}$. If the branch terminates at the reset-price floor, then $\Pi_{T-1} \to 0$ and $Z_{T-1} \to \underline{Z}$ as $\theta \downarrow \theta_C^*$, and Equation (6) gives $S_{T-1} \to (1-\theta)\underline{Z}^{-\epsilon} = (1-\theta)^{-1/(\epsilon-1)}$. Then $S_{T-1}/(1-\theta+\theta S_{T-1}) = 1 + \theta/(\epsilon-1) + O(\theta^2)$, so $1 - m = \zeta|\bar{x}| - \varphi\theta/(\epsilon-1) + O(\bar{x}^2, \theta^2, \theta|\bar{x}|)$. Equating $\eta(\theta)(1-m)$ with $\theta/(\epsilon-1) + O(\theta^2)$ gives the approximation for θ_C^*. □

Proof of Proposition 3. With $\sigma = 1$, $Y_t^{1-\sigma} = 1$ for all t, so $V_{2,T} = \sum_{h \geq 0} (\beta\theta)^h = 1/(1-\beta\theta)$ and Equation (14) gives $\eta(\theta) = [1 + \beta_{T-1}\theta/(1-\beta\theta)]^{-1}$. Hence $G(\theta) = \eta(\theta)(1-\bar{m}(\theta)) - [1 - (1-\theta)^{1/(\epsilon-1)}]$ with $\bar{m}(\theta)$ as in Equation (21), and $S_{T-2} \leq B(\theta)$ and $\Pi_{T-1} \leq 1$ imply $m \leq \bar{m}(\theta)$. An equilibrium with $\Pi_{T-1} \leq 1$ therefore satisfies $G(\theta) \leq \eta(\theta)(1-m) - [1 - (1-\theta)^{1/(\epsilon-1)}] < 0$ by Inequality (19), and no such equilibrium exists when $G(\theta) \geq 0$. The function G is continuous on $(0,1)$ with $G(\theta) \to 1 - e^{\zeta \bar{x}} > 0$ as $\theta \to 0$. As $\theta \to 1$, $\bar{m}(\theta) \to e^{\zeta \bar{x}}$ and $\eta(\theta) \to (1-\beta)/(1-\beta+\beta_{T-1})$, so the first term of G tends to a value below one while the bracket tends to one, and G has a negative limit. Hence the set in Equation (24) is non-empty and $\underline{\theta} > 0$. By the definition of the infimum, $G(\theta) \geq 0$ for all $\theta < \underline{\theta}$, so no equilibrium with $\Pi_{T-1} \leq 1$ exists there. The deflationary branch has solutions at values of θ just above θ_C^*, so $\theta_C^* \geq \underline{\theta}$. For the approximation, $\beta_{T-1} = 1 + O(|\bar{x}|)$ gives $\eta(\theta) = 1 - \theta + O(\theta^2, \theta|\bar{x}|)$. Also $B(\theta) = 1 + \epsilon\theta/(\epsilon-1) + O(\theta^2)$ and $\theta(B(\theta)-1) = O(\theta^2)$, so $B/(1-\theta+\theta B) = 1 + \epsilon\theta/(\epsilon-1) + O(\theta^2)$ and $1 - \bar{m}(\theta) = \zeta|\bar{x}| - \varphi\epsilon\theta/(\epsilon-1) + O(\bar{x}^2, \theta^2, \theta|\bar{x}|)$. With $1 - (1-\theta)^{1/(\epsilon-1)} = \theta/(\epsilon-1) + O(\theta^2)$,

$$G(\theta) = \zeta|\bar{x}| - \frac{(1+\epsilon\varphi)\,\theta}{\epsilon-1} + O(\bar{x}^2, \theta^2, \theta|\bar{x}|),$$

and setting $G(\theta) = 0$ gives Equation (25). □

Proof of Proposition 4. For $\Pi \in (0, 1]$, $(\Pi - 1)^2 < 1$, so $m_R(\Pi) \leq e^{\zeta \bar{x}}(1 - \psi/2)^{-\varphi}$, while $\psi(\Pi - 1)\Pi \geq -\psi/4$. Hence $F(\Pi) \geq -\psi/4 + (\epsilon - 1)[1 - e^{\zeta \bar{x}}(1 - \psi/2)^{-\varphi}] > 0$ under the condition in (a), and there is no root. For (b), $F(1) = (\epsilon - 1)(1 - e^{\zeta \bar{x}}) > 0$. If $e^{\zeta \bar{x}}(1 - \psi/2)^{-\varphi} > 1$, then $\lim_{\Pi \to 0} F(\Pi) = -(\epsilon - 1)[e^{\zeta \bar{x}}(1 - \psi/2)^{-\varphi} - 1] < 0$. If instead $\psi/4 > (\epsilon - 1)(1 - e^{\zeta \bar{x}})$, then $F(1/2) \leq -\psi/4 + (\epsilon - 1)(1 - e^{\zeta \bar{x}}) < 0$, since $m_R(1/2) \geq e^{\zeta \bar{x}}$. In either case continuity gives a root in $(0, 1)$. For the approximation, the first condition in (b) reads $\varphi\psi/2 \gtrsim \zeta|\bar{x}|$, and the condition in (a) reads $\psi < 4(\epsilon - 1)\zeta|\bar{x}|/[2(\epsilon - 1)\varphi + 1]$. Both give $\psi^* \approx 2\zeta|\bar{x}|/\varphi$ when $2(\epsilon - 1)\varphi \gg 1$, and $\psi \approx (\epsilon - 1)\theta$ for small θ. □

Proof of Proposition 5. Fix $T \geq 1$ and $\theta \in \Theta_{T+1}$, and take an equilibrium with a trap of length $T + 1$. With firm-specific labor markets, $C_t = Y_t$ and the equilibrium conditions are the Euler equation (2), the price index (5) and the reset condition (8) with its recursions. None of these involves price dispersion, because $MC_t^s = \chi Y_t^{\zeta}$ does not. Under Rotemberg pricing the equilibrium conditions are the Euler equation (2), Equation (9) and the resource constraint $C_t = k_t Y_t$. In both cases the conditions at date t involve only variables at t and $t+1$, the discount factor β_t and the nominal rate i_t. For every variable define the shifted sequence $v'_t \equiv v_{t+1}$, $t \geq 0$. Its discount factors $\beta'_t = \beta_{t+1}$ equal $1/(1 + \bar{r})$ for $t \leq T - 1$ and β for $t \geq T$, its nominal rate is zero for $t \leq T - 1$, and $\Pi'_t = 1$ for $t \geq T$. The shifted sequence therefore satisfies every equilibrium condition of a trap of length T, so $\theta \in \Theta_T$. Hence $\Theta_{T+1} \subseteq \Theta_T$. The infimum of a subset is at least the infimum of the set, and if Θ_{T+1} is empty then $\theta^*(T + 1) = 1 \geq \theta^*(T)$. In both cases $\theta^*(T + 1) \geq \theta^*(T)$. □

C Numerical Methods

All models are solved in levels, without approximation, in double precision, and the terminal regime of Lemma 1 is imposed exactly from T on.

Calvo, firm-specific labor. Allocations do not depend on dispersion, so the model is solved by backward recursion from T: given $(\Pi_{t+1}, Y_{t+1}, V^I_{1,t+1}, V_{2,t+1})$, the Euler equation gives Y_t, the recursions give $V^I_{1,t}$ and $V_{2,t}$, Equation (8) gives Z_t, and Equation (5) gives Π_t. Non-existence is recorded when $Z_t \leq \underline{Z}(\theta)$ at some t.

Calvo, economy-wide labor. Dispersion is a backward-looking state that feeds into marginal cost, so the model is solved as a system of T equations in u_t, with $Z_t = \underline{Z}(\theta) + e^{u_t}$ so that Lemma 2 holds by construction. Given $\{u_t\}$, Equation (5) gives $\{\Pi_t\}$, Equation (6) gives $\{S_t\}$ forward from $S_{-1} = 1$, the terminal regime gives Y_T and $V_{2,T}$ (the post-exit sums run over the infinite horizon and are computed exactly, in closed form when $\sigma = 1$ and otherwise with a closed-form geometric tail once dispersion equals one to machine precision), and the Euler equation and Equation (7) give residuals $\log Z_t - \log[\frac{\epsilon}{\epsilon-1} V_{1,t}/V_{2,t}]$. The system is solved by a Powell hybrid method with continuation in θ from 0.99 downward in steps of 0.002, refined to 10^{-7} near the existence boundary, with a residual tolerance of 10^{-10}. Continuation in θ stalls slightly before the existence boundary because Π_0 falls very fast there. From the last continuation point we therefore fix $\log \Pi_0$ along a sequence of values down to -10 and solve for θ and $u_1, \ldots, u_{T-1}$. The resulting θ is the reported threshold. Stopping the sequence at $\log \Pi_0 = -3$ instead of -10 changes the threshold by less than 10^{-7}.

Rotemberg. The model is solved by backward recursion. Given period-$(t+1)$ variables, the Euler equation gives C_t, and Equation (9) becomes a scalar equation in $L_t \equiv \log k_t$, where $k_t = 1 - \frac{\psi}{2}(\Pi_t - 1)^2 = C_t/Y_t$ and $\Pi_t \in (0, 1]$ is increasing in L_t. Working with L_t rather than Π_t keeps solutions accurate when adjustment costs absorb almost all output: along long traps with $\psi > 2$ the root can lie within 10^{-14} of $1 - \sqrt{2/\psi}$, or closer, where a grid in Π_t cannot separate it from the pole at $k_t = 0$. We keep the largest solution in L_t, which is the solution with the mildest deflation because Π_t is increasing in L_t. At the baseline calibration, the equation has at most one solution at every date for all trap lengths and natural rates in Table 1, on a grid of θ with step 0.005. Taking the largest solution guards against spurious sign changes that rounding error creates where Π_t is numerically zero. To find all solutions, we evaluate the residual of this equation on a grid of values of L_t. Wherever the residual changes sign between two neighboring grid points, a solution lies between them, and we locate it by repeatedly halving that interval (bisection). The grid has extra points near $L_t = 0$, where deflation is mild, and near the lower end of the domain, where adjustment costs absorb almost all output. A grid can miss two solutions that lie close together between the same two grid points, because the residual then does not change sign there. To catch such pairs, we take each point where the residual comes closest to zero without crossing it and refine it with a numerical optimizer, which tests whether the residual crosses zero nearby. Non-existence is recorded when no root exists, or when the

root would require $L_t < -700$. At $T = 8$ the solution was validated against an independent 60-digit computation.

Thresholds. The thresholds for the firm-specific Calvo and Rotemberg models are computed by bisection in θ, which assumes that existence holds on an interval of θ.

Computational validation. On a grid of θ with step 0.005, existence holds on a single interval for the firm-specific Calvo and Rotemberg models at $T = 1$ and $T = 8$, and for the economy-wide Calvo model at $T = 1$. With $T = 1$ the price index (5), the dispersion equation (6) and the Euler equation (2) give every other variable as a function of the reset price Z_0. The model then reduces to one equation in Z_0, the reset condition (7). We evaluate the residual of this equation on a fine grid of values of Z_0 above the reset-price floor and locate every sign change, which finds all equilibria. For the economy-wide model with $T > 1$ this validation is not available, and the threshold refers to the end of the deflationary branch followed from high θ. The Rotemberg threshold at $T = 8$ is identical with grids of 1,001, 2,001 and 8,001 points. The economy-wide threshold at $T = 8$ is unchanged to seven digits when the continuation step is halved or the residual tolerance is tightened to 10^{-12}.

The replication script (`replication_calvo_zlb.py`) reproduces all tables and computational validations.